

Temporal Variations in Emotional Tone in Reddit Discussions: A Python-Based VAD Analysis

Vladyslava Pedan
LMT, Adam Mickiewicz University, Poznań, Poland
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Abstract

This project examines whether emotional tone in Reddit relationship discussions varies across different times of day. The study uses a balanced dataset of 2,000 Reddit relationship_advice submissions, divided equally into morning, afternoon, evening, and night groups. Emotional tone was measured with the NRC Valence, Arousal, and Dominance (VAD) Lexicon, with a specific focus on valence and arousal. Valence was used to estimate the positive or negative tone of each text, while arousal was used to estimate emotional intensity. The analysis was completed through a Python pipeline that included data loading, text cleaning, tokenization, lexicon-based scoring, statistical testing, visualization, and SQLite storage. The initial hypotheses predicted that nighttime posts would show lower valence and that evening or nighttime posts would show higher arousal. The results did not fully support these expectations. Valence differences across time groups were small and not statistically significant, while arousal differences were statistically significant but strongest in the morning rather than in the evening or night. These findings suggest that emotional tone in Reddit relationship discussions is not simply determined by time of day and that emotional intensity may appear in unexpected temporal patterns.

1. Introduction

Social media has become an important source of real-life language data because people use online platforms to describe experiences, ask for advice, express emotions, and react to personal problems. Unlike surveys or prepared texts, social media posts are usually written naturally and often include timestamps. This makes them useful for studying not only what people write, but also when certain types of emotional language appear. For a Python-based text analysis project, this type of data is especially valuable because it can be collected, cleaned, grouped, measured, and visualized through a clear computational workflow.

This project focuses on Reddit relationship_advice discussions. This community is relevant for emotional analysis because users often write about personal conflict, uncertainty, anxiety, sadness, stress, and difficult relationship decisions. Many posts are longer than typical short social media messages, so they provide enough text for lexicon-based emotional scoring. At the same time, the topic of the subreddit gives the analysis a clear emotional context, because relationship problems often involve both negative feelings and strong emotional intensity.

The main aim of this project is to examine whether emotional tone in Reddit relationship discussions changes depending on the time of day. The assumption behind this question is that emotional expression may vary across daily time periods. For example, nighttime posts might be expected to

sound more negative or emotionally intense because users may be tired, isolated, or reflecting on personal problems. Morning posts may also contain strong emotional language if users write after conflict, stress, or lack of sleep. Therefore, the relationship between time of day and emotional tone should be tested with data rather than assumed.

The project compares four time-of-day groups: morning, afternoon, evening, and night. Emotional tone is measured through two dimensions: valence and arousal. Valence shows whether a text is more positive or negative, while arousal shows how emotionally intense the text is. This makes it possible to study emotional tone in a more detailed way than a simple positive-negative classification.

The research question is:

Does the emotional tone of Reddit relationship discussions vary depending on the time of day?

Two hypotheses were tested.

H1: Nighttime texts will have lower valence, meaning more negative emotional tone.

H2: Evening and nighttime texts will have higher arousal, meaning stronger emotional intensity.

2. Background

Sentiment analysis is a field of computational text analysis that studies opinions, attitudes, emotions, and subjectivity in written language. Pang and Lee (2008) and Liu (2012) describe sentiment analysis as a way to automatically identify positive, negative, and more complex emotional meanings in text. This is useful because large collections of online posts cannot be analyzed manually in an efficient and consistent way. Computational methods make it possible to compare many texts using the same procedure.

However, emotional meaning is more complex than a simple positive or negative label. A text can be negative but calm, or negative and highly emotional. For example, two Reddit posts may both describe a relationship problem, but one may sound sad and reflective while another may sound urgent, angry, or emotionally intense. Because of this, a dimensional approach to emotion is useful for this project. It allows the analysis to separate emotional polarity from emotional intensity.

The Valence, Arousal, and Dominance model describes emotion through several dimensions. Valence measures how positive or negative a word is. Arousal measures the level of emotional intensity or activation. Dominance measures the sense of control or power connected with a word. In this project, the focus is mainly on valence and arousal because they are most directly related to the research question. Valence helps show whether posts are more positive or negative, while arousal helps show whether posts are emotionally intense.

The project uses the NRC Valence, Arousal, and Dominance Lexicon developed by Mohammad (2018). The lexicon provides human ratings for English words on the three VAD dimensions, which makes it suitable for a transparent lexicon-based analysis. Mohammad (2020) also explains that emotions in text can be studied through affective dimensions, not only through basic sentiment

polarity. This is important for the current project because the goal is not only to ask whether Reddit posts are negative, but also whether they differ in emotional intensity.

This project was also inspired by research on temporal mood patterns in social media. Lampos, Preotiuc-Pietro, Cohn, and Aletras (2013) used time-stamped Twitter content to study mood patterns and showed that online language can be analyzed across time. Their work demonstrates that timestamps can add an important layer to sentiment and emotion analysis. My project follows a similar general idea, but applies it to Reddit relationship_advice posts and uses a smaller Python-based workflow.

The present project does not aim to make broad claims about all social media users. Instead, it focuses on one specific online community and tests whether emotional tone changes across broad parts of the day. This narrower focus makes the project more manageable while still allowing a meaningful analysis of temporal emotional patterns in online relationship discussions.

3. Methodology

The project follows a complete Python-based data analysis pipeline. The main stages of the work were data acquisition, data cleaning, text preprocessing, VAD scoring, statistical analysis, visualization, and data storage. This structure was used to make the analysis organized, reproducible, and easy to check through separate output files.

The dataset was created from Reddit relationship_advice submissions. The source data came from the HuggingFaceGECLM REDDIT_submissions dataset, which includes a relationship_advice split. I created a balanced working sample with 2,000 texts: 500 morning posts, 500 afternoon posts, 500 evening posts, and 500 night posts. Balanced sampling was used because the goal was to compare emotional tone across groups equally, not to measure the natural posting frequency of Reddit users. This means that each time-of-day group had the same weight in the final comparison.

The time groups were defined as morning (6:00–11:59), afternoon (12:00–17:59), evening (18:00–23:59), and night (00:00–5:59). These categories were used to compare emotional tone across broad parts of the day. The timestamps were processed as UTC timestamps, which made the grouping technically consistent, although it also created a limitation because UTC time may not always match the local time of Reddit users.

Before emotional scoring, the text data was cleaned and standardized. The Python script loaded the CSV file, normalized the columns, created text fields, and converted timestamps. The title and body of each Reddit post were combined into one text field so that the emotional score would represent the full content of the post rather than only the title. Then the text was converted to lowercase. Links, unnecessary symbols, punctuation noise, and extra spaces were removed using regular expressions. After that, the text was tokenized into individual words. Very short texts were not useful for emotional analysis, so the program used a minimum word requirement before scoring. This reduced the chance that a post would receive an emotional score based on too little lexical evidence.

The emotional scoring was based on the NRC Valence, Arousal, and Dominance Lexicon developed by Mohammad (2018). This lexicon contains human ratings for English words on three affective

dimensions: valence, arousal, and dominance. The scores range from 0 to 1. In this project, I used only valence and arousal because these two dimensions were most relevant to the research question. Valence was used to estimate whether a text had a more positive or negative emotional tone, while arousal was used to estimate emotional intensity.

For each Reddit post, the cleaned text was tokenized into individual words. The program then checked which tokens appeared in the NRC VAD Lexicon. Words that were found in the lexicon received their corresponding valence and arousal scores. After that, the script calculated the average valence and average arousal for each post based on all matched words. A higher valence score indicates a more positive tone, while a lower valence score indicates a more negative tone. A higher arousal score indicates stronger emotional intensity. This approach follows the general idea described by Mohammad (2020), who explains that affective states in text can be studied not only through positive or negative sentiment, but also through more detailed emotional dimensions.

After scoring the texts, the script grouped the results by time of day: morning, afternoon, evening, and night. For each group, it calculated the number of texts, mean valence, mean arousal, standard deviations, average word count, and VAD coverage. VAD coverage was important because it showed how many words in each text could actually be matched with the lexicon. This helped evaluate whether the emotional scores were based on enough lexical evidence.

To test whether the observed differences across time groups were statistically meaningful, one-way ANOVA and the Kruskal-Wallis test were used. ANOVA was used to compare mean emotional scores across the four groups, while Kruskal-Wallis was used as a non-parametric alternative because text-based emotional scores may not perfectly follow a normal distribution. A p-value below 0.05 was treated as statistically significant. The final project outputs included cleaned CSV files, VAD scoring results, group summaries, statistical test results, a SQLite database, interpretation notes, and five visualizations.

4. Results

The final dataset contained 2,000 Reddit relationship_advice posts. The data was balanced across four time-of-day groups, with 500 posts in each group. This balanced structure made it possible to compare morning, afternoon, evening, and night posts under equal sample conditions.

Table 1. Mean emotional tone by time of day.

Time of Day	N	Mean Valence	SD Valence	Mean Arousal	SD Arousal
Morning	500	0.618	0.077	0.464	0.054
Afternoon	500	0.624	0.067	0.455	0.050
Evening	500	0.625	0.066	0.456	0.052
Night	500	0.619	0.070	0.463	0.058

Table 1 shows that mean valence was very close across all four time groups. The lowest mean valence was found in the morning group, with a score of 0.618. The night group was almost the same, with a score of 0.619. Afternoon and evening showed slightly higher valence scores, with evening having the highest mean valence at 0.625.

Arousal showed a slightly clearer difference between groups. The highest mean arousal was found in the morning group, with a score of 0.464. The night group was very close, with a score of 0.463. Afternoon and evening had lower mean arousal scores.

Table 2. Statistical test results for emotional tone by time of day.

Metric	Test	Statistic	p-value	Interpretation
Valence	One-way ANOVA	1.187	0.313	Not significant
Valence	Kruskal-Wallis	3.056	0.383	Not significant
Arousal	One-way ANOVA	4.143	0.006	Significant
Arousal	Kruskal-Wallis	11.147	0.011	Significant

Table 2 shows different statistical patterns for valence and arousal. For valence, the one-way ANOVA produced $p = 0.313$, and the Kruskal-Wallis test produced $p = 0.383$. Since both p-values are above 0.05, the differences in valence across time groups were not statistically significant.

For arousal, the one-way ANOVA produced $p = 0.006$, and the Kruskal-Wallis test produced $p = 0.011$. Since both p-values are below 0.05, the differences in arousal across time groups were statistically significant.

Overall, the results show that valence remained relatively stable across the four time groups, while arousal showed statistically significant variation.

5. Visualizations

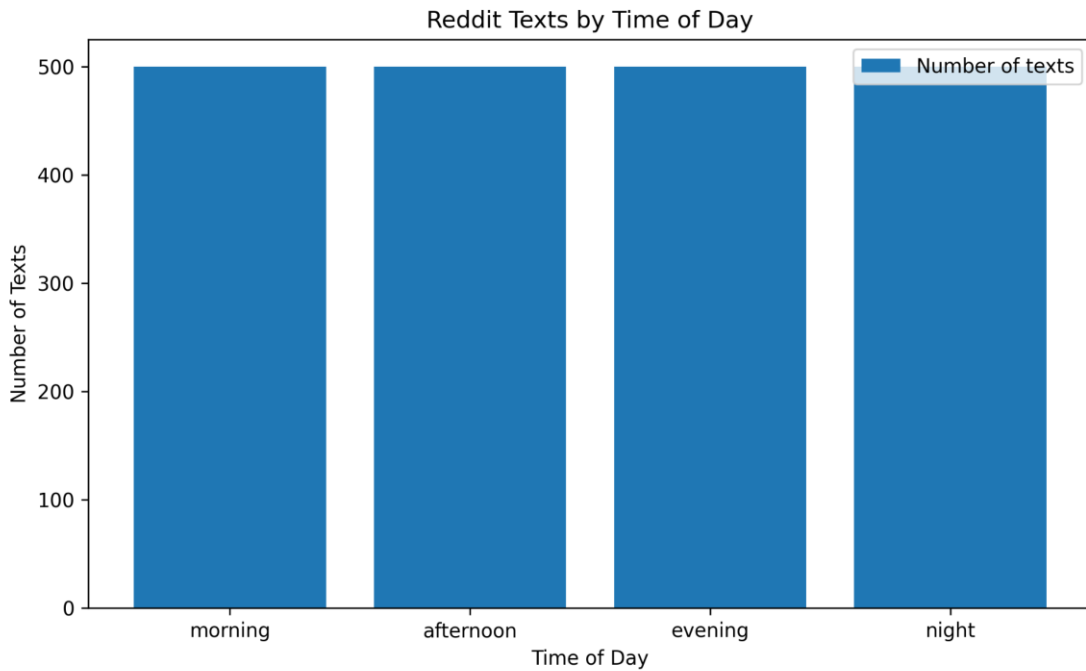


Figure 1. Balanced sample of Reddit relationship_advice texts across four time-of-day groups.

Figure 1 presents the number of Reddit posts in each time-of-day group. The chart shows that the dataset was intentionally balanced, with 500 posts in morning, afternoon, evening, and night. This balance makes the comparison between the groups more equal. However, it also means that the figure should not be interpreted as the natural posting frequency of Reddit users across the day.

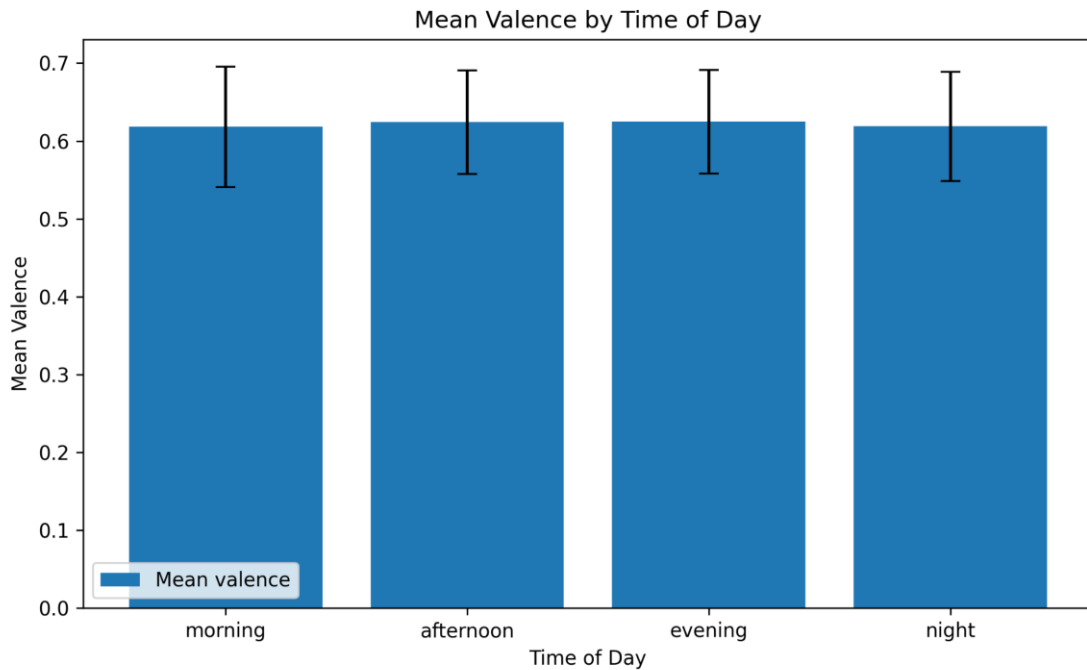


Figure 2. Mean valence by time of day. Lower values indicate more negative emotional tone.

Figure 2 presents the mean valence score for each time-of-day group. The values are close across all four groups. Morning has the lowest mean valence, while evening has the highest mean valence. The small distance between the bars visually supports the statistical result that valence differences were not significant.

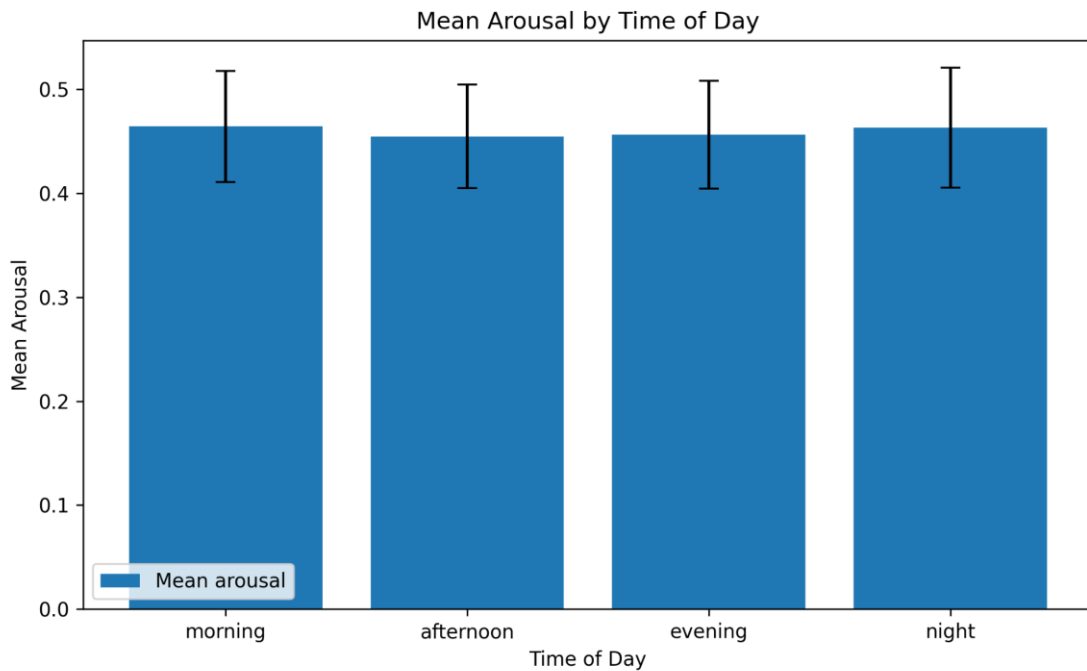


Figure 3. Mean arousal by time of day. Higher values indicate stronger emotional intensity.

Figure 3 presents the mean arousal score for each time-of-day group. Arousal is highest in the morning and almost as high at night. Afternoon and evening show lower arousal scores. This chart makes the arousal pattern easier to see than the table alone.

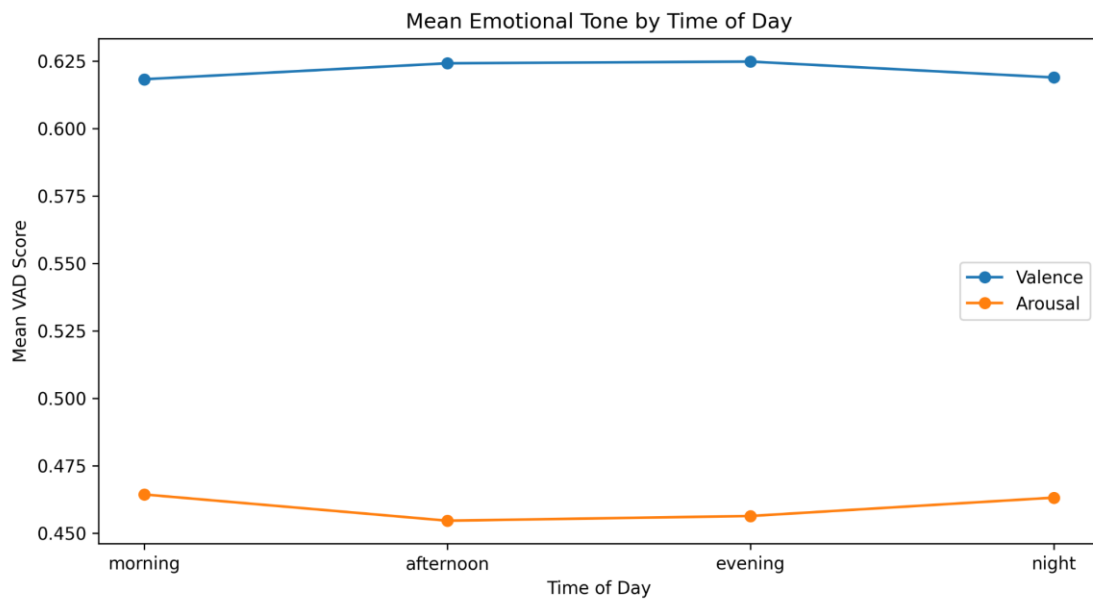


Figure 4. Mean emotional tone by time-of-day group, comparing valence and arousal.

Figure 4 compares valence and arousal across the four time-of-day groups in one chart. Valence remains higher than arousal in every group, but the two measures follow different patterns. The chart shows that arousal varies more visibly across time groups than valence.

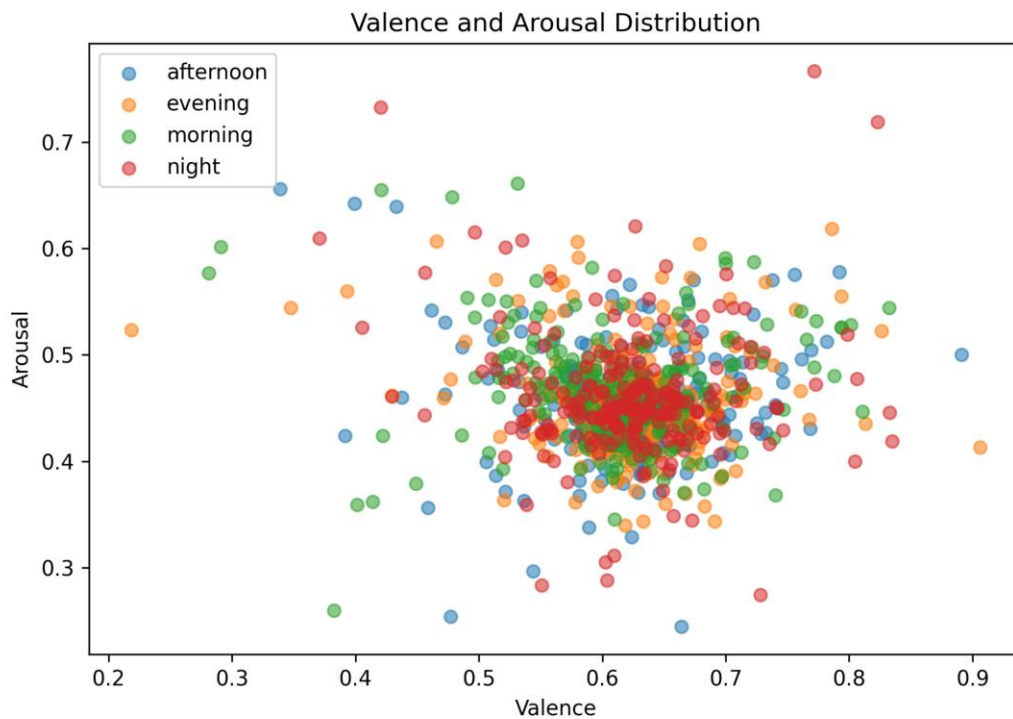


Figure 5. Scatter plot of valence and arousal scores by time-of-day group.

Figure 5 shows the relationship between valence and arousal for individual Reddit posts. The points from different time groups overlap strongly. This indicates that emotional tone varies within each group and that time of day does not clearly separate posts into distinct emotional categories.

6. Discussion

The results show that emotional tone in Reddit relationship_advice discussions does not follow a simple time-of-day pattern. The first hypothesis predicted that nighttime texts would have lower valence, meaning a more negative emotional tone. This hypothesis was not supported. Although the night group had a relatively low mean valence score, the morning group had the lowest score overall. More importantly, the differences between the four groups were small, and the statistical tests showed that valence differences were not significant. This suggests that the positive-negative tone of relationship advice posts remained relatively stable across morning, afternoon, evening, and night.

The second hypothesis predicted that evening and nighttime texts would have higher arousal, meaning stronger emotional intensity. This hypothesis was only partially supported. Night posts had relatively high arousal, but the highest mean arousal was found in the morning group. This result is important because it shows that emotional intensity was not limited to later parts of the day. Instead, emotionally intense language appeared in both morning and night posts, while afternoon and evening posts had slightly lower arousal scores.

One possible explanation is that relationship problems may become emotionally intense at different moments of the day. Morning posts may be written after users have experienced conflict, reflected on a problem overnight, or woken up still feeling stressed about a relationship issue. Night posts may also be emotionally intense because users may write when they feel tired, isolated, or unable to resolve a personal problem. In this sense, the results suggest that emotional intensity in relationship_advice discussions may be connected less to one specific time period and more to the personal situation that motivates users to post.

Another important explanation is methodological. Reddit timestamps are stored in UTC, so the assigned time-of-day group may not represent the user's actual local time. For example, a post classified as morning in UTC could have been written at night in the user's local time zone. This limitation is especially important for Reddit because users may come from many different countries and time zones. Therefore, the results should be interpreted as patterns based on UTC posting time, not as exact patterns of users' local daily routines.

The findings also show the value of separating valence and arousal. If the project had only measured positive versus negative tone, the main conclusion would be that time of day did not matter very much. However, arousal showed statistically significant variation across the groups. This suggests that time of day may be more relevant to emotional intensity than to emotional positivity or negativity. In other words, the posts were not necessarily more negative at particular times, but they did differ in how emotionally intense they appeared.

There are several limitations to the project. First, as mentioned above, UTC timestamps may not match local user time. Second, the sample was intentionally balanced, with 500 posts in each group. This was useful for comparing groups equally, but it means that the analysis does not show natural Reddit posting frequency. Third, the VAD lexicon scores words individually, so it may miss sarcasm, negation, context, and complex sentence meaning. For example, a post may contain positive words while describing a negative situation. Fourth, the project focuses only on the relationship_advice

subreddit, so the findings should not be generalized to all Reddit communities or all forms of online emotional expression.

Future work could improve the analysis in several ways. A larger dataset could be used to check whether the same pattern appears with more posts. Local time estimation could improve the time-of-day grouping if user location information were available. A future version could also compare the lexicon-based VAD method with machine learning or transformer-based sentiment models. This would make it possible to test whether more context-aware methods produce similar results.

Overall, the project shows that emotional patterns in social media text can be more complex than the original hypotheses suggest. The results did not support the idea that nighttime posts are clearly more negative, but they did show that arousal varies significantly across time groups. The strongest conclusion is that valence remained stable, while emotional intensity showed statistically significant but unexpected temporal variation. From a technical perspective, the project also demonstrates a complete Python workflow for social media text analysis, including data acquisition, cleaning, VAD scoring, statistical comparison, visualization, and database storage.

7. Resources

This project used both data resources and Python tools to complete the analysis. The main text data came from the HuggingFaceGECLM REDDIT_submissions dataset, specifically the relationship_advice split. This dataset was used to create the balanced sample of Reddit posts for the four time-of-day groups.

The emotional scoring was based on the NRC Valence, Arousal, and Dominance Lexicon. This lexicon was used to assign valence and arousal scores to words in the Reddit posts. These scores were then averaged at the post level and summarized by time group.

The project was completed in Python. The main libraries and modules used were pandas for reading and organizing CSV files, numpy for numerical operations, matplotlib for visualizations, sqlite3 for database storage, argparse for command-line input, re for regular-expression-based text cleaning, and scipy for statistical testing.

The project produced several output files. These included the raw Reddit data, cleaned and processed text data, VAD scoring results, group summary tables, hourly summary tables, statistical test results, five visualizations, a SQLite database, figure captions, and interpretation notes. Together, these files show the full workflow from dataset preparation to final analysis.

8. Conclusion

This project analyzed emotional tone in Reddit relationship_advice discussions across four time-of-day groups: morning, afternoon, evening, and night. Using Python and the NRC VAD Lexicon, the analysis measured two affective dimensions: valence and arousal. Valence was used to examine positive-negative emotional tone, while arousal was used to examine emotional intensity.

The results did not support the hypothesis that nighttime posts would be clearly more negative. Morning had the lowest mean valence, but the difference between time groups was small and not

statistically significant. This suggests that positive-negative emotional tone remained relatively stable across the four groups.

The results for arousal were more meaningful. Arousal differed significantly across time groups, but the highest mean arousal was found in the morning rather than in the evening or night. This means that the second hypothesis was only partially supported. Night posts were emotionally intense, but they were not the highest group.

Overall, the findings show that emotional patterns in Reddit relationship discussions are more complex than the original expectations suggested. Time of day did not strongly affect valence, but it was more clearly related to emotional intensity. The project also demonstrates how Python can be used to build a complete research pipeline for social media text analysis, including data acquisition, cleaning, lexicon-based scoring, statistical testing, visualization, and database storage.

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